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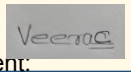
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	SLOT C/2025	**Date:**	03-10-2026

Code of Conduct

I C Veera Bharadwaj Kumar(192565061) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept	**Q1, Q4**

	sequence, expected utility, rational versus omniscient agent	

Annexure A – Analytical Problem Solving

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes Three

AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.

- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47

Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	**20%**	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of	**15%**	Clearly interprets results with units and	Basic interpretation with	Limited or unclear interpretation	No interpretation

Faculty In-charge Signature: _____ Date: _____

Course Coordinator Signature: _____ Date: _____

Program Director Signature: _____ Date: _____

Results		engineering relevance	minor omissions		

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Question 1: Two-Room Vacuum World

Given Information

- Environment: two squares **A and B**
- Initial state: **A = Dirty, B = Dirty**
- Agent starts at **A**
- Actions: **Suck, Right, Left, NoOp**
- Lifetime: **T = 6 time steps**
- Performance:
  - **+1 for every clean square** at the end of each time step
  - **−p** for every Left/Right movement
  - Given **p = 1** unless otherwise specified
- Dirt never reappears.

For every step:

$$\text{Step Score} = (\text{Number of clean squares}) - (\text{Move penalty})$$

where move penalty is p if the action is Left/Right and 0 otherwise.

(a) PEAS Description — [4 Marks]

PEAS stands for **Performance measure, Environment, Actuators, Sensors**.

PEAS Component	Description
P – Performance Measure	+1 for each clean square at the end of every time step; −p for every Left/Right movement
E – Environment	Two squares A and B; initially both are Dirty; dirt does not reappear
A – Actuators	Suck, Left, Right, NoOp
S – Sensors	Sensor perceives the agent's current location (A or B) and whether the current square is Dirty or Clean

Therefore:

$$P = +1 \text{ per clean square} - p \text{ per movement}$$

The agent must balance **cleaning squares** with **avoiding unnecessary movement**.

(b) Trace Agent X — [5 Marks]

Agent X: Simple Reflex Agent

Rules:

1. If current square is **Dirty** → **Suck**
2. If current square is **Clean** → move to the other square

Initial state:

A=D, B=D

Agent starts at A.

Step-by-step trace

Time	Location	Percept	Action	Clean squares at end	Move penalty	Step score for p=1
1	A	Dirty	**Suck**	1	0	1-0=1
2	A	Clean	**Right**	2	1	2-1=1
3	B	Clean	**Left**	2	1	2-1=1
4	A	Clean	**Right**	2	1	2-1=1
5	B	Clean	**Left**	2	1	2-1=1
6	A	Clean	**Right**	2	1	2-1=1

Explanation

At time 1, X cleans A.

At time 2, X moves from A to B. Now both A and B are clean.

However, X has **no memory**. Whenever it sees a clean square, its rule says:

"If Clean → move to the other square."

Therefore, it keeps moving:

even though both squares are already clean.

Total score of X for p=1

$S_X=1+1+1+1+1+1$

$S_X=6$

(c) Trace Agent Y — [4 Marks]

Agent Y: Reflex Agent with Internal State

Rules:

1. If current square is **Dirty** → Suck
2. If current square is Clean and it remembers both squares are clean → **NoOp**
3. Otherwise → move to the other square

Agent Y has memory, so after cleaning/observing both squares, it remembers that **both are clean**.

Step-by-step trace

Time	**Location**	**Percept**	**Action**	**Clean squares at end**	**Move penalty**	**Step score for \(p=1\)**
1	A	Dirty	**Suck**	1	0	1-0=1
2	A	Clean	**Right**	2	1	2-1=1
3	B	Clean	**NoOp**	2	0	2-0=2
4	B	Clean	**NoOp**	2	0	2-0=2
5	B	Clean	**NoOp**	2	0	2-0=2
6	B	Clean	**NoOp**	2	0	2-0=2

Total score of Y

S_Y=1+1+2+2+2+2

S_Y=10

Thus, for p=1:

X=6,Y=10

(d) Total Score as a Function of \(p\) — [3 Marks]

Agent X

Agent X performs:

- 1 Suck operation
- 5 movements

At time 1:

Score=1

For times 2–6, both squares are clean:

Score per step=2-p

Therefore,

$S_X=1+5(2-p)$

$S_X=11-5p$

Agent Y

Agent Y performs:

- 1 Suck
- 1 movement
- 4 NoOp actions

Therefore:

$S_Y=1+(2-p)+4(2)$

$S_Y=1+2-p+8$

$S_Y=11-p$

Compare X and Y

Y is strictly better when:

$S_Y > S_X$

Substitute:

$11-p > 11-5p$

Subtract 11:

$-p > -5p$

Add 5p to both sides:

$4p > 0$

Therefore:

$p > 0$

Results

Value of p	**Comparison**
$p>0$	Y>X
$p=0$	Y=X=11
$p<0$	X>Y mathematically

Since p represents a **penalty**, normally . Hence, for any positive movement penalty:

Agent Y is strictly better than Agent X

(e) Which Agent Is Rational? — [4 Marks]

Under the given performance measure, **Agent Y is rational for p>0** because it achieves the same cleaning objective while avoiding unnecessary movement penalties.

Why is Y rational?

After time 2:

A=Clean, B=Clean

Agent Y uses its internal state to remember this fact. It therefore chooses:

NoOp

instead of repeatedly moving between the two squares.

This gives Y:

$S_Y=11-p$

while X gets:

$S_X=11-5p$

For $p>0$:

$S_Y>S_X$

Why can't X match Y?

The important difference is **internal state/memory**.

Agent X is a **simple reflex agent**. Its decision depends only on its **current percept**:

Current location + current dirt status

Once both squares are clean, X repeatedly receives percepts such as:

- At A: **A, Clean**
- At B: **B, Clean**

Its rule is always:

If Clean → Move to the other square.

Therefore:

- **A, Clean→Right**
- **B, Clean→Left**

and it continues moving.

Agent Y, however, maintains information about the **percept history**. It knows:

"I have already cleaned/visited both squares, and neither can become dirty again."

Therefore, it can select **NoOp**.

Key point about rationality

Rationality depends on:

1. **The performance measure** — unnecessary movement has a penalty.
2. **The percept history** — previous observations can tell the agent that both squares are already clean.
3. **The agent's internal state** — Y can use this history, while X cannot.

So, for p>0:

Agent Y is rational under the given performance measure

while X is less effective because its simple reflex rule cannot use past percepts.

Question 2: Task Environment Analysis and Agent Selection

(a) Classification of E1, E2 and E3 on all six properties

The six environment properties are:

1. Observability
2. Number of agents
3. Determinism
4. Episodic / Sequential
5. Static / Dynamic
6. Discrete / Continuous

The convention given in the question is that the harder value scores 1 and the easier value scores 0. Semi-dynamic scores 0.5.

E1: Autonomous Taxi Driving Through City Traffic

1. Observability: Partially Observable

Justification: The taxi cannot observe everything in the environment, such as hidden vehicles, pedestrians behind obstacles, or the intentions of other drivers.

2. Number of agents: Multi-agent

Justification: Other drivers, pedestrians and vehicles also make decisions that affect the taxi.

3. Determinism: Stochastic

Justification: Traffic movements, pedestrian behaviour, weather and actions of other drivers are uncertain.

4. Episodic / Sequential: Sequential

Justification: Each driving decision affects future positions, safety, route and travel time.

5. Static / Dynamic: Dynamic

Justification: The traffic environment changes continuously while the taxi is sensing and deciding.

6. Discrete / Continuous: Continuous

Justification: Vehicle positions, speeds, steering angles and time change continuously.

Therefore:

E1 = Partially Observable, Multi-agent, Stochastic, Sequential, Dynamic, Continuous.

E2: Part-Picking Robot on a Moving Conveyor Belt

1. Observability: Partially Observable

Justification: The robot uses a camera, but the camera may not provide complete information because of occlusion, lighting, object orientation and sensing limitations.

2. Number of agents: Single agent

Justification: The task can be performed by one robot, and no other decision-making agent is specified.

3. Determinism: Stochastic

Justification: Camera sensing errors, variations in object position and movement, and uncertainty in robot gripping make the outcome uncertain.

4. Episodic / Sequential: Sequential

Justification: The robot's actions occur repeatedly, and the current picking operation affects the remaining parts and future actions.

5. Static / Dynamic: Dynamic

Justification: The conveyor belt is moving, so the positions of parts change while the robot is sensing and acting.

6. Discrete / Continuous: Continuous

Justification: The positions, velocities and movements of the conveyor, parts and robot arm vary continuously.

Therefore:

E2 = Partially Observable, Single-agent, Stochastic, Sequential, Dynamic, Continuous.

E3: Chess-Playing Agent in a Tournament Game with a Clock

1. Observability: Fully Observable

Justification: The complete chessboard position and all legally relevant pieces are visible to both players.

2. Number of agents: Multi-agent

Justification: Two players, the chess agent and its opponent, make decisions that affect the game.

3. Determinism: Deterministic

Justification: Given a board position and a legal move, the resulting board position is completely determined by the rules of chess.

4. Episodic / Sequential: Sequential

Justification: Every move changes the board and affects the possible moves and outcomes of later moves.

5. Static / Dynamic: Semi-dynamic

Justification: The chessboard does not change while the agent is thinking, but the tournament clock continues to run and affects performance.

Therefore:

E3 = Fully Observable, Multi-agent, Deterministic, Sequential, Semi-dynamic, Discrete.

SUMMARY TABLE

Property	**E1: Autonomous Taxi**	**E2: Part-Picking Robot**	**E3: Chess Agent**
Observability	Partially Observable	Partially Observable	Fully Observable
Number of agents	Multi-agent	Single-agent	Multi-agent
Determinism	Stochastic	Stochastic	Deterministic
Episodic / Sequential	Sequential	Sequential	Sequential
Static / Dynamic	Dynamic	Dynamic	Semi-dynamic
Discrete / Continuous	Continuous	Continuous	Discrete

(b) Difficulty Score and Ranking

According to the question:

- Easier value = 0
- Harder value = 1
- Semi-dynamic = 0.5

The scores for each property are calculated as follows.

E1: Autonomous Taxi

Observability: Partially Observable = 1

Number of agents: Multi-agent = 1

Determinism: Stochastic = 1

Episodic / Sequential: Sequential = 1

Static / Dynamic: Dynamic = 1

Discrete / Continuous: Continuous = 1

Therefore:

Difficulty Score of E1

= 1 + 1 + 1 + 1 + 1 + 1

= 6

Therefore,

E1 = 6 points

E2: Part-Picking Robot

Observability: Partially Observable = 1

Number of agents: Single-agent = 0

Determinism: Stochastic = 1

Episodic / Sequential: Sequential = 1

Static / Dynamic: Dynamic = 1

Discrete / Continuous: Continuous = 1

Therefore:

Difficulty Score of E2

= 1 + 0 + 1 + 1 + 1 + 1

= 5

Therefore,

E2 = 5 points

E3: Chess Agent

Observability: Fully Observable = 0

Number of agents: Multi-agent = 1

Determinism: Deterministic = 0

Episodic / Sequential: Sequential = 1

Static / Dynamic: Semi-dynamic = 0.5

Discrete / Continuous: Discrete = 0

Therefore:

Difficulty Score of E3

= 0 + 1 + 0 + 1 + 0.5 + 0

= 2.5

Therefore,

E3 = 2.5 points

Difficulty Ranking

Rank	**Environment**	**Difficulty Score**
1	E1 – Autonomous Taxi	6
2	E2 – Part-Picking Robot	5
3	E3 – Chess Agent	2.5

Thus, according to the given scoring convention:

Hardest: E1 – Autonomous Taxi

Next: E2 – Part-Picking Robot

Easiest: E3 – Chess Agent

(c) Suitable Agent Architecture

1. Hardest Environment: E1 – Autonomous Taxi

The most suitable architecture is:

UTILITY-BASED AGENT

Reason:

An autonomous taxi operates in a partially observable, multi-agent, stochastic, sequential, dynamic and continuous environment.

The taxi has to consider several competing factors, such as:

- Safety
- Travel time
- Distance
- Traffic conditions
- Passenger comfort
- Fuel or energy consumption
- Actions of other road users

A utility-based agent can assign utility values to different possible actions and select an action that provides the most desirable overall outcome.

For example, when choosing between two routes, the agent can consider:

Route 1: Shorter but heavy traffic

Route 2: Longer but safer and less congested

A utility-based agent can consider the different consequences rather than simply following one fixed rule.

Therefore:

E1 → Utility-Based Agent

2. Easiest Environment: E3 – Chess

The most suitable architecture is:

GOAL-BASED AGENT

Reason:

Chess is fully observable and deterministic. The agent can see the complete board and knows the rules of the game.

The main goal is to achieve a winning state, such as checkmate.

The agent can therefore:

1. Observe the current board.
2. Generate legal moves.
3. Predict possible future positions.
4. Select moves that help achieve the goal of checkmate.

Unlike a simple reflex agent, a goal-based agent can consider future consequences of its actions.

The tournament clock makes the environment semi-dynamic, but the main state of the chessboard is discrete and fully observable.

Therefore:

E3 → Goal-Based Agent

(d) Interpretation

- The complexity of an agent program depends strongly on the properties of its environment, not simply on the name or domain of the task.
- For example, autonomous taxi driving is complex because the environment is partially observable, stochastic, multi-agent, sequential, dynamic and continuous. The agent must maintain information about the environment, deal with uncertainty, predict other agents' behavior and make decisions continuously.
- In contrast, chess is fully observable and deterministic. The board has a discrete state and the rules completely determine the result of a move. Therefore, the agent does not need to deal with the same level of uncertainty as an autonomous taxi.
- Similarly, the part-picking robot has to handle partial camera observations, uncertainty, continuous motion and a changing conveyor belt.

Thus, environment properties determine the information and reasoning required by the agent.

The general relationship is:

Environment properties → Required agent capabilities → Complexity of agent program

For example:

- Partial observability → Need for internal state/model
- Stochastic environment → Need to handle uncertainty
- Multi-agent environment → Need to consider other agents
- Sequential environment → Need to consider future consequences
- Dynamic environment → Need continuous monitoring and fast decisions
- Continuous environment → Need continuous sensing and control

Therefore, a more challenging environment generally requires a more sophisticated agent architecture

Question 3: Turing Test Data Across the History of AI

Given:

- Total interrogations for each system = 50
- Correctly identified = number of times the judge correctly identified the machine
- Fooling rate = percentage of times the machine was mistaken for a human
- Turing's criterion = average interrogator should have no more than 70% chance of correctly identifying the machine.
- Guessing level = 50%

Formula:

Correct-identification rate = (Correctly identified / Total interrogations) × 100

Fooling rate = ((Total interrogations - Correctly identified) / Total interrogations) × 100

(a) Correct-identification rate and fooling rate

1. ELIZA

Total interrogations = 50

Correctly identified = 47

Correct-identification rate:

= (47 / 50) × 100

= 94%

Therefore,

Correct-identification rate = 94%

Number of times fooled:

= 50 - 47

= 3

Fooling rate:

= (3 / 50) × 100

= 6%

Therefore,

ELIZA:
Correct-identification rate = 94%
Fooling rate = 6%

2. Chatbot B

Total interrogations = 50

Correctly identified = 36

Correct-identification rate:

= (36 / 50) × 100

= 72%

Number of times fooled:

= 50 - 36

= 14

Fooling rate:

= (14 / 50) × 100

= 28%

Therefore,

Chatbot B:
Correct-identification rate = 72%
Fooling rate = 28%

3. System C

Total interrogations = 50

Correctly identified = 29

Correct-identification rate:

= (29 / 50) × 100

= 58%

Number of times fooled:

= 50 - 29

= 21

Fooling rate:

= (21 / 50) × 100

= 42%

Therefore,

System C:

Correct-identification rate = 58%

Fooling rate = 42%

Summary Table:

System	Correctly Identified	Correct Rate	Fooled	Fooling Rate
ELIZA	47/50	94%	3/50	6%
Chatbot B	36/50	72%	14/50	28%
System C	29/50	58%	21/50	42%

(b) Turing's 70% criterion and distance from 50% chance level

Turing's statement says that an average interrogator would have no more than a 70% chance of making the correct identification after five minutes.

Therefore, we use:

Correct-identification rate <= 70%

as the criterion.

Now check each system.

1. ELIZA

Correct-identification rate = 94%

94% > 70%

Therefore, ELIZA does NOT satisfy Turing's 70% criterion.

Distance from 50% chance level:

= 94% - 50%

= 44 percentage points

Therefore, ELIZA is 44 percentage points above the guessing level.

2. Chatbot B

Correct-identification rate = 72%

72% > 70%

Therefore, Chatbot B does NOT satisfy Turing's 70% criterion.

Distance from 50% chance level:

= 72% - 50%

= 22 percentage points

Therefore, Chatbot B is 22 percentage points above the guessing level.

3. System C

Correct-identification rate = 58%

58% <= 70%

Therefore, System C satisfies Turing's 70% criterion.

Distance from 50% chance level:

= 58% - 50%

= 8 percentage points

Therefore, System C is 8 percentage points above the guessing level.

Summary:

System	Correct Rate	70% Criterion	Distance from 50%
ELIZA	94%	Does NOT satisfy	44 percentage points
Chatbot B	72%	Does NOT satisfy	22 percentage points
System C	58%	Satisfies	8 percentage points

Interpretation:

System C has the lowest correct-identification rate, 58%. This means judges were closest to the 50% guessing level for System C. Its fooling rate is 42%, meaning it was taken for a human in 42% of the interrogations.

(c) Improvement in fooling rate

Fooling rates:

ELIZA = 6%

Chatbot B = 28%

System C = 42%

1. Improvement from ELIZA to System C

Percentage-point improvement:

= Fooling rate of System C - Fooling rate of ELIZA

= 42% - 6%

= 36 percentage points

Therefore:

Improvement = 36 percentage points

Ratio:

= Fooling rate of System C / Fooling rate of ELIZA

= 42 / 6

= 7

Therefore:

System C's fooling rate is 7 times the fooling rate of ELIZA.

If expressed as percentage increase:

= ((42 - 6) / 6) × 100

= (36 / 6) × 100

= 600%

Therefore, the fooling rate increased by 600%, which is equivalent to becoming 7 times as large.

2. Improvement from Chatbot B to System C

Percentage-point improvement:

= 42% - 28%

= 14 percentage points

Therefore:

Improvement from Chatbot B to System C = 14 percentage points.

Summary:

Comparison Improvement

ELIZA → System C 36 percentage points
ELIZA → System C 7 times the fooling rate
ELIZA → System C 600% increase
Chatbot B → System C 14 percentage points

(d) Interpretation: Acting Humanly vs Thinking/Rationality

The Turing Test measures whether a machine can "act humanly" during a conversation. It does not directly measure whether the machine is actually thinking like a human or acting rationally.

System C has a fooling rate of 42%. This means that in 42% of the interrogations, the judges incorrectly identified the machine as a human.

However, this result alone does not prove that System C thinks like a human.

One important idea from the foundations of AI is the Chinese Room argument.

The Chinese Room argument, proposed by John Searle, argues that a system can produce appropriate responses to language without necessarily understanding the meaning of the language. A system may follow rules or manipulate symbols and still appear to understand a conversation.

Therefore, even if System C successfully convinces some judges that it is human, the result only provides evidence about its ability to imitate human conversation. It does not by itself prove:

1. That System C has human-like understanding.
2. That System C has consciousness.
3. That System C reasons in the same way as humans.
4. That System C always chooses rational actions.

Thus:

Turing Test → Measures ability to act humanly.

It does NOT by itself prove genuine human-like thinking or rationality.

Question 4: Rational Route Choice under Uncertainty

Given:

Utility formula:

U = 100 - T - 20L

where:

T = travel time in minutes

L = 1 if the taxi arrives after 55 minutes

L = 0 if the taxi arrives within 55 minutes

Deadline = 55 minutes

Therefore:

If T <= 55, then L = 0

U = 100 - T

If T > 55, then L = 1

U = 100 - T - 20

(a) Utility of Each Route in Each Outcome

1. Route R1: Highway

Normal outcome:

Travel time = 30 minutes

Since 30 <= 55:

L = 0

U = 100 - 30 - 20(0)

U = 70

Therefore:

R1 Normal Utility = 70

Disrupted outcome:

Travel time = 70 minutes

Since 70 > 55:

L = 1

U = 100 - 70 - 20(1)

U = 100 - 70 - 20

U = 10

Therefore:

R1 Disrupted Utility = 10

This trip is LATE.

2. Route R2: City Road

Normal outcome:

Travel time = 45 minutes

Since 45 <= 55:

L = 0

U = 100 - 45 - 20(0)

U = 55

Therefore:

R2 Normal Utility = 55

Disrupted outcome:

Travel time = 60 minutes

Since 60 > 55:

L = 1

U = 100 - 60 - 20(1)

U = 20

Therefore:

R2 Disrupted Utility = 20

This trip is LATE.

3. Route R3: Ring Road

Normal outcome:

Travel time = 50 minutes

Since 50 <= 55:

L = 0

U = 100 - 50 - 20(0)

U = 50

R3 is never disrupted.

Therefore:

R3 Utility = 50

This trip is ON TIME.

Utility Table

Route	Outcome	Travel Time	L	Utility	Status
R1: Highway	Normal	30 min	0	70	On time
R1: Highway	Disrupted	70 min	1	10	Late
R2: City Road	Normal	45 min	0	55	On time
R2: City Road	Disrupted	60 min	1	20	Late
R3: Ring Road	Normal	50 min	0	50	On time

(b) Expected Utility and Rational Choice

Expected utility is calculated as:

Expected Utility = Σ [Probability of outcome × Utility of outcome]

1. Expected Utility of R1

R1 has:

Probability of normal = 0.70

Probability of disruption = 0.30

Utilities:

Normal = 70

Disrupted = 10

Therefore:

EU(R1) = (0.70 × 70) + (0.30 × 10)

EU(R1) = 49 + 3

EU(R1) = 52

Therefore:

Expected Utility of R1 = 52

2. Expected Utility of R2

R2 has:

Probability of normal = 0.90

Probability of disruption = 0.10

Utilities:

Normal = 55

Disrupted = 20

Therefore:

EU(R2) = (0.90 × 55) + (0.10 × 20)

EU(R2) = 49.5 + 2

EU(R2) = 51.5

Therefore:

Expected Utility of R2 = 51.5

3. Expected Utility of R3

R3 is never disrupted.

Probability of normal = 1

Utility = 50

Therefore:

EU(R3) = 1 × 50

EU(R3) = 50

Therefore:

Expected Utility of R3 = 50

Expected Utility Summary

Route	**Expected Utility**
R1: Highway	52
R2: City Road	51.5
R3: Ring Road	50

The rational choice is the route with the highest expected utility.

Highest expected utility = 52

Therefore:

Rational choice = R1: Highway

(c) New Percept: Highway Accident Alert

A new radio alert reports a major accident on the highway.

The probability of disruption on R1 increases from:

0.30 to 0.80

Therefore:

Probability of normal R1 = 1 - 0.80 = 0.20

Utilities remain:

Normal R1 = 70

Disrupted R1 = 10

Expected Utility of R1:

EU(R1) = (0.20 × 70) + (0.80 × 10)

EU(R1) = 14 + 8

EU(R1) = 22

Therefore:

Expected Utility of R1 = 22

The other routes still have:

R2 = 51.5

R3 = 50

Comparison:

Route	**Expected Utility**
R1: Highway	22
R2: City Road	51.5
R3: Ring Road	50

The highest expected utility is 51.5.

Therefore, after receiving the new percept, the rational agent should choose:

R2: City Road

(d) Actual Utility and Rationality

In part (b), the rational agent chose R1 because:

EU(R1) = 52

which was higher than:

EU(R2) = 51.5

and:

EU(R3) = 50

Suppose the disruption actually occurred on R1.

Then:

Travel time = 70 minutes

Since 70 > 55:

L = 1

Actual utility:

U = 100 - 70 - 20(1)

U = 10

Therefore:

Actual utility obtained = 10

Was the agent irrational?

No.

The agent was NOT irrational.

At the time of making the decision, the agent had a 30% probability of disruption on R1 and no information about the future accident beyond that probability.

The agent selected R1 because it had the highest expected utility:

R1 = 52

R2 = 51.5

R3 = 50

The fact that the unfavorable outcome actually occurred does not make the original decision irrational.

Rationality is judged based on the information available to the agent when the decision was made, not only by looking at the outcome afterward.

Rational Agent vs Omniscient Agent

A rational agent:

- Uses the percepts and information currently available.
- Chooses the action expected to maximize performance.
- Cannot know future events with certainty.

An omniscient agent:

- Would know exactly what will happen in the future.
- Would know whether the highway accident would occur.
- Could choose the route that gives the best actual outcome with perfect knowledge.

Therefore:

A rational agent does not need to make the best decision in every actual outcome.

It needs to make the best decision based on the information available at that time.

(e) Interpretation

Rationality depends on both the percept sequence and the performance measure.

First, the percept sequence provides information about the current environment. In part (b), the agent knows that R1 has a 30% probability of disruption. Based on this information, R1 has the highest expected utility.

In part (c), a new percept arrives: a radio alert reports a major accident on the highway. This changes the probability of disruption on R1 from 30% to 80%.

The utilities of the roads themselves have not changed:

R1 Normal = 70

R1 Disrupted = 10

However, the probability of the outcomes has changed. Therefore, the expected utility changes from:

EU(R1) = 52

to:

EU(R1) = 22

As a result, the rational decision changes from R1 to R2.

This demonstrates that a rational agent continuously uses new percepts to update its decision.

The performance measure also affects the decision. Here, arriving after 55 minutes causes an additional penalty of 20 utility units. Therefore, the possibility of being late significantly affects the expected utility of a route.